

Postprocessing and plotting of simulink

this form of script gives us better control of how many plot do we wanna draw at one time.

Postprocessing

Data extracted and calculated. Ready to plot.

Simulation Parameters Used

Design Configuration File: design_params\design_param1.m

Maximum Cavern Pressure (p_store_max): 100.0 bar

Cavern Volume (V_cavern): 5.00e+06 m³

TES Volume (V_tes): 50000.0 m³

TES Mass (m_tes): 1.080e+08 kg

Max TES Temperature (T_tes_max): 499.9 °C

Expansion Inlet Temperature (T_expand): 99.9 °C

Compressor Efficiency (eta_comp): 83.3%

Expander Efficiency (eta_exp): 82.0%

Transmission Efficiency (eta_tran): 97.0%

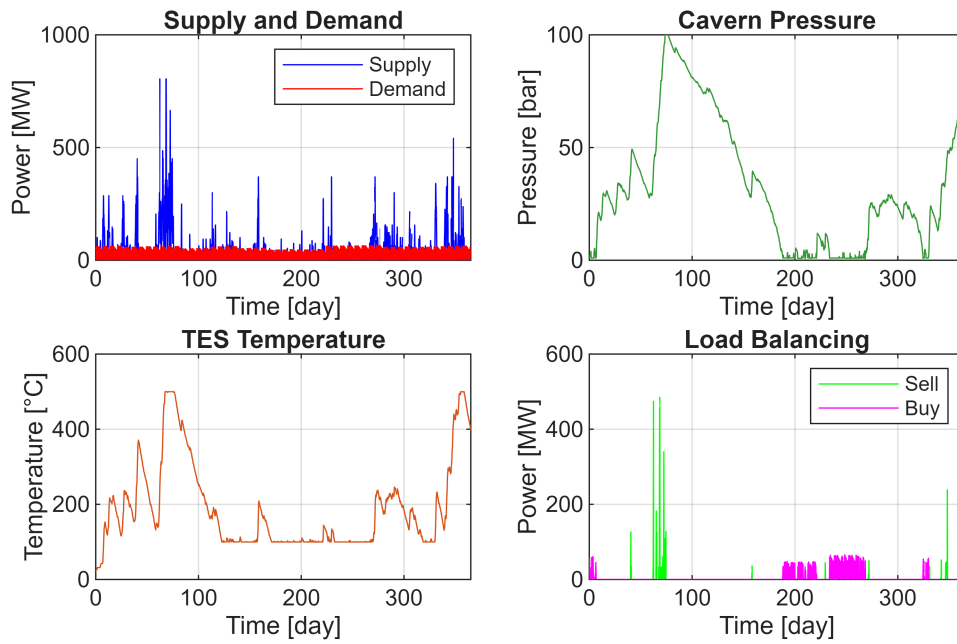
Pipe Diameter (D_pipe): 50.00 m

Flow Velocity (v_flow): 2000.0 m/s

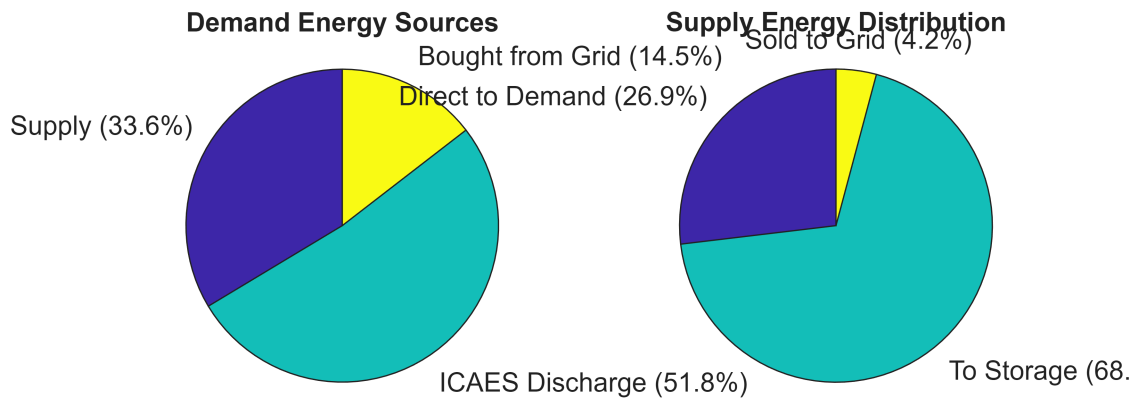
Grid Charging Power Limit (P_limit): 300.0 MW

Polytropic Index (n_poly): 1.14

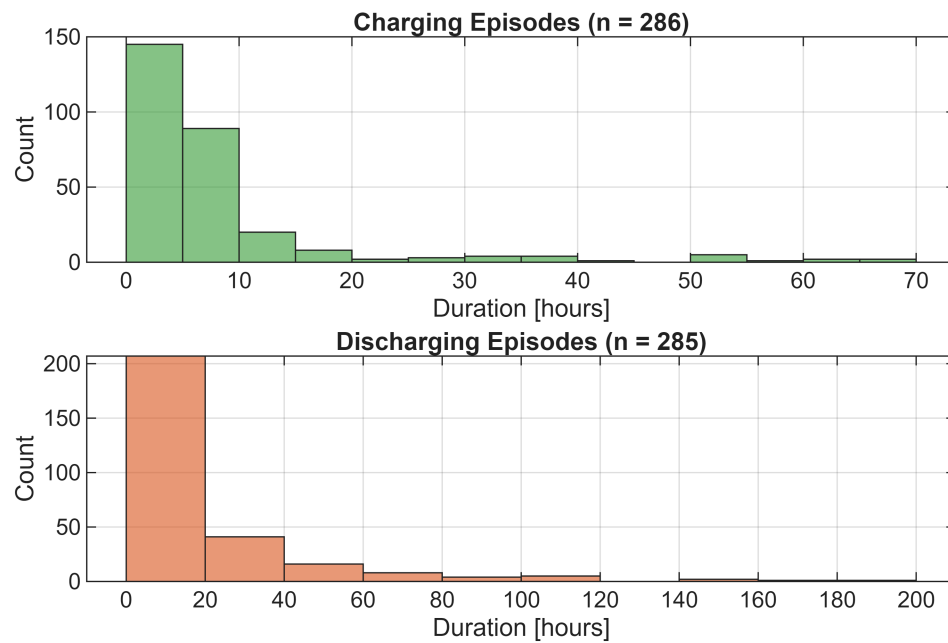
Plot: System State Overview (Supply/Demand, Pressure, TES Temp, Buy/Sell)



Plot: Energy Pie Charts (Demand Sources & Supply Distribution)



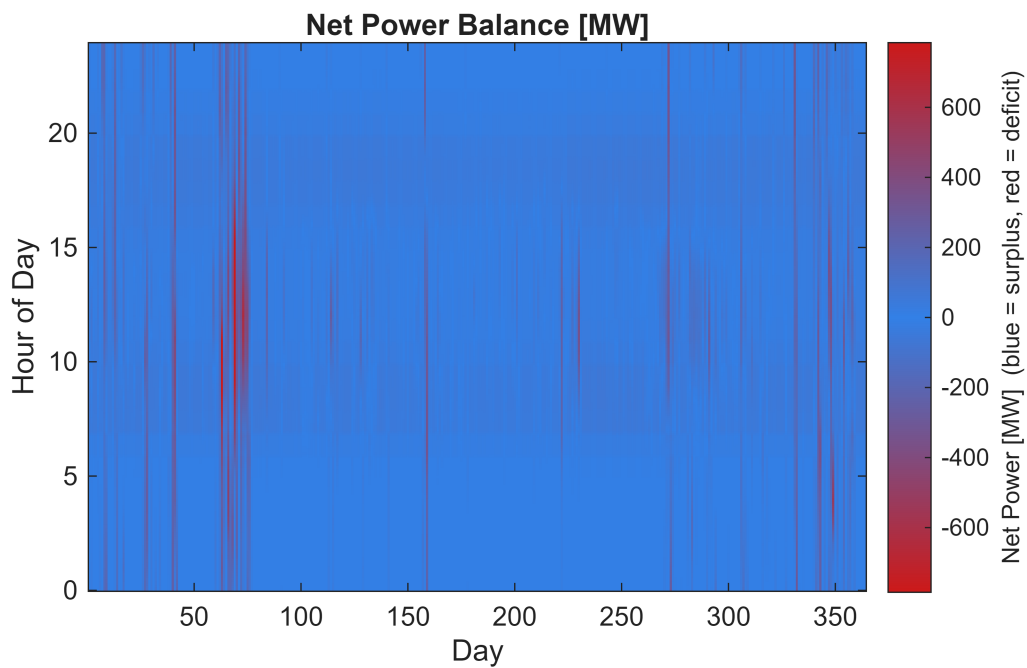
Charging/Discharging Duration Histogram



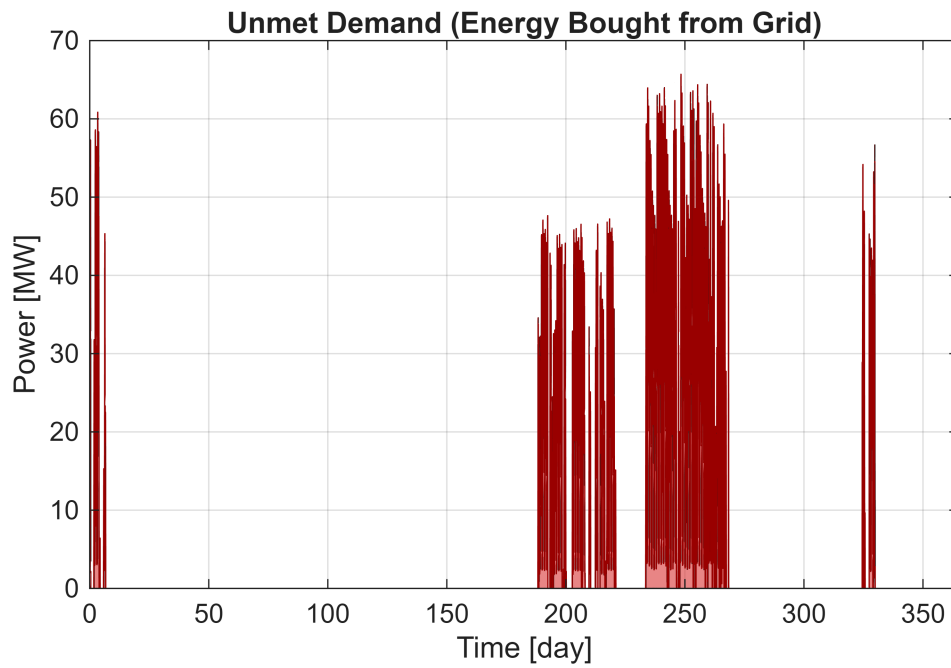
Charging – median: 4.8 h, max: 67.6 h, total episodes: 286

Discharging – median: 6.2 h, max: 185.1 h, total episodes: 285

Surplus/Deficit Heatmap (Hour-of-Day vs Day)



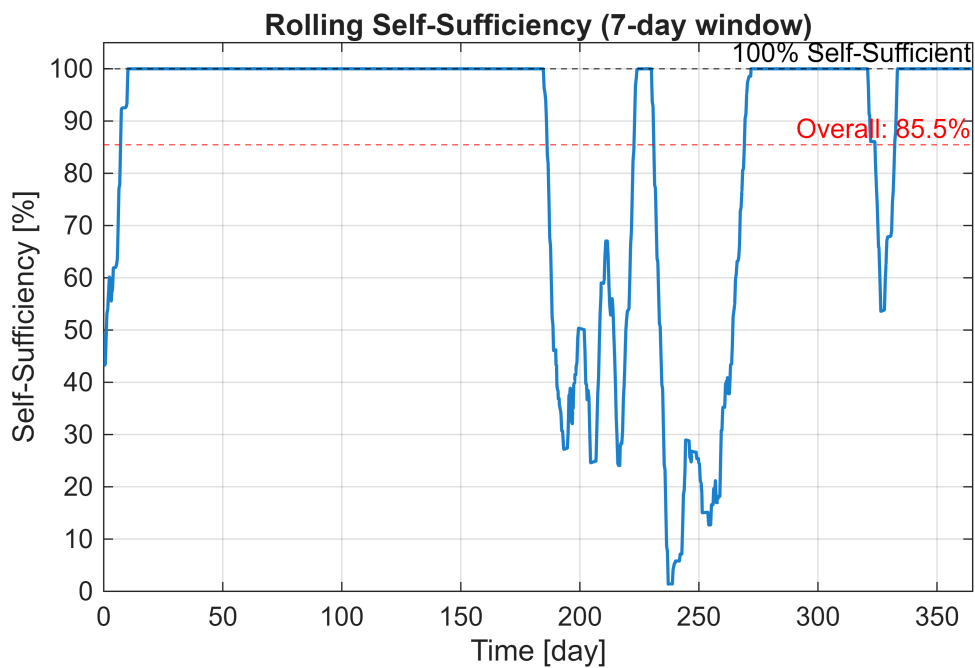
Unmet Demand



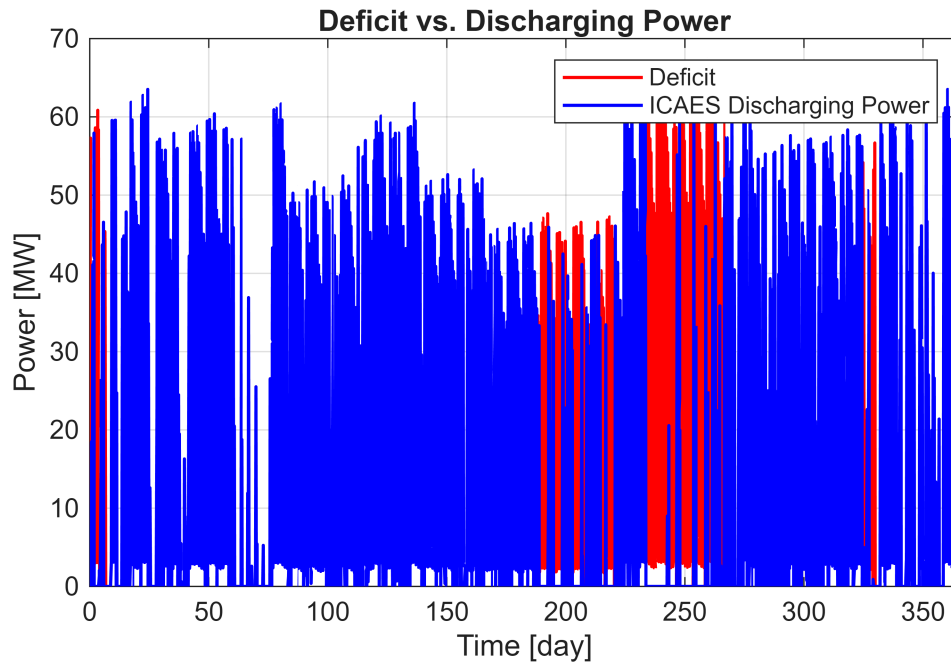
Unmet demand events: 16701 of 105118 timesteps (15.9% of time)

Peak unmet demand: 65.7 MW

Rolling Self-Sufficiency



Plot: Deficit vs. ICAES Discharging Power



(replaces your existing efficiency block)

EST Efficiency + Loss Breakdown

Total surplus available : 2305675783052.00

Energy discharged (ICAES): 1308074561785.00

Round-trip efficiency : 56.7%

Self-sufficiency ratio : 85.5%

